Giacomo Cappelletto

Software Engineer and D1 Crew Athlete

Email: giacomo.cappelletto@icloud.com
Mobile: +39 324 681 4985
linkedin.com/in/giacomo-cappelletto/
github.com/JJCAPPE

Professional Summary

Computer Engineering student balancing competitive rowing for Boston University Men's Crew with a passion for software development and AI. The discipline required for international-level athletics translates directly to my approach in engineering, whether optimizing application performance or tackling complex technical challenges. I'm driven by the opportunity to create meaningful software solutions that solve real problems for people. Experience spans full-stack development, mobile applications, and machine learning, with a strong foundation in modern frameworks and cloud technologies. Always looking to learn and contribute to impactful projects.

Education

- **Boston University, Boston, MA** 2024 – 2028
  B.Sc. in Computer Engineering; GPA: 3.96
- **H-Farm International School, Venice, Italy** 2022 – 2024
  International Baccalaureate Diploma Programme *Higher Levels in Mathematics, Physics, and Computer Science*

Experience

- **SOCIETA CAPPELLETTO S.R.L., Treviso, Italy** May 2025 – Sep 2025
  Full Stack Native App Developer – Tauri Inventory Management System Rebuild
  - **Migration & Performance**: Rebuilt Electron-based inventory system using Tauri-React-Rust, reducing app size by 70% and improving startup time by 60%. Implemented parallel API processing and Promise.all() concurrency patterns, achieving 43% faster search performance (3.7s to 2.09s).
  - **Advanced Features**: Developed comprehensive modification history tracking with Firebase integration, real-time inventory reconciliation across locations, and GraphQL-optimized Shopify API interactions. Added SKU-based instant search with concurrent fallback mechanisms.
  - **Operational Impact**: Enhanced system reliability and reduced memory footprint by 80% through native Rust backend. Improved user workflow efficiency with immediate inventory updates and eliminated search delays, supporting seamless multi-location inventory management.
- **TickIT GmbH, Boston, MA (Remote)** Oct 2024 – Jan 2025
  Frontend & Backend Engineer – Ticketing Platform Startup
  - **Overview**: Engineered responsive user interfaces using Next.js, Shadcn components, and Tailwind CSS; developed robust backend functionality with Ruby on Rails and PostgreSQL, ensuring scalable API integrations.
  - **Contribution**: Provided key architectural insights on ticket authentication flow and streamlined Agile development processes while rapidly mastering new frameworks.
- **SOCIETA CAPPELLETTO S.R.L., Treviso, Italy** May 2024 – Jul 2024
  Full Stack Native App Developer – Online Inventory Management System
  - **Project**: Developed an Electron-React-Node.js application integrating the Shopify API and Firebase Firestore to dynamically update online shop inventory across two locations.
  - **Impact**: Eliminated online inventory discrepancies and reduced manual labor times at checkout by 80%, significantly improving operational efficiency.
- **H-Farm International School, Treviso, Italy** Mar 2024 – Apr 2024
  Python Developer – Student Meeting Application Developer
  - **Application**: Designed and implemented a Python Streamlit application to leverage school timetables from Managebac, automating meeting request emails using local LLMs in Ollama.
  - **Collaboration**: Collaborated closely with faculty to tailor the solution, ensuring seamless integration with existing systems and improved scheduling efficiency.
- **DLF Treviso, Treviso, Italy** Sep 2023 – Nov 2023
  iOS Developer – Rowing LogBook Application

- **Development**: Designed and developed an iOS app using Swift/SwiftUI (with Swift Charts) to enable athletes to log training sessions and analyze performance data.
- **Process**: Managed full-cycle development from planning and design to implementation and evaluation, incorporating direct user feedback for continuous improvement.

- **E-TRASH, Treviso, Italy** Feb 2023 – Jun 2023
  CAD Engineer Intern
  - **CAD Modeling**: Designed and developed the first prototype of an intelligent trash bin using Autodesk Fusion 360, focusing on efficiency and structural integrity.
  - **Optimization**: Applied topology optimization and load testing to enhance the prototype's durability and weight distribution.
  - **3D Rendering**: Created high-quality 3D visualizations in Blender for presentation and product refinement.

Projects

- **NoteWorthy – Automated Note Conversion Platform**: Led development of noteworthyy.xyz a Next.js/TypeScript application that converts handwritten notes into styled PDFs and LaTeX code. Integrated the Google AI Studio API for content processing, implemented GitHub/Google OAuth for secure authentication, containerized a Node.js LaTeX compiler with Docker, and deployed via Google Cloud Run on Vercel.
- **CIFAR-10 Image Recognition with Neural Architecture Search**: Engineered a machine learning solution using TensorFlow/Keras to perform image recognition on the CIFAR-10 dataset. Developed a neural architecture search pipeline that generated and evaluated CNN architectures with convolutional layers, max-pooling, and global average pooling. Automated candidate training, performance logging, and selection of the best-performing model for extended training and deployment.

Skills

- **Languages**: JavaScript (JSX/TSX, JS), Python, Ruby, C/C++, Swift/SwiftUI, Java
  Frameworks & Libraries: Next.js, React.js, Ruby on Rails, Node.js, Electron, SwiftUI
  Technologies: Tailwind CSS, PostgreSQL, Docker, Firebase, Shopify API, Streamlit, Ollama, Google Cloud Run, Github
  Methodologies: Agile Development, API Integration, Cloud Deployment